

Harry Nguyen

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PROFESSIONAL SUMMARY

CS student at Georgia State (May 2027) with production experience at Chrome scale and solo AI platform ownership. At Google, shipped TypeScript/React developer tooling and C++ IPC to 3B+ users. Built and operate TiMoto AI end-to-end - an AI agent memory platform using vLLM, gRPC, and session-context persistence at sub-50ms p99. Open source contributor to Pulumi's developer tools Go CLI for multi-cloud IaC. Experienced in agentic application development, TypeScript API design, and production platform engineering. Case study: timoto.ai.

CORE COMPETENCIES

Agentic Application Development

TypeScript / API Design

Developer Tools & Open Source

AI Agent Memory & Session Context

Production ML Serving (vLLM)

Distributed Systems (gRPC / Protobuf)

Platform Engineering (AWS / Terraform)

Low-Latency Backend (C++ / Go)

WORK EXPERIENCE

Google

May 2025 -- Aug 2025

Software Engineer Intern - Chrome Browser

- Architected event-driven **TypeScript/React** system with observer pattern decoupling UI state propagation - accelerated developer feature delivery **68%** across **25K+ lines** of Chromium codebase at **95% test coverage**
- Designed **C++ IPC transport layer** with Protocol Buffers - selected Protobuf over custom serialization for schema evolution and cross-language compatibility; shipped to Chrome stable serving **3B+ active users** at **sub-50ms p99**, **10K+ req/sec**
- Identified settings navigation as p99 bottleneck at 1,200ms; designed **lock-free concurrent trie search** eliminating mutex contention - **96% latency reduction**, zero production regressions
- Collaborated with Chrome infrastructure team on design documents and code reviews at 95% test coverage; changes adopted into production branch reviewed by senior Chrome engineers

Develop for Good

May 2024 -- Aug 2024

Software Engineer Intern

- Designed stateless BaaS on AWS - selected JWT over session auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+ on large datasets; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

CoderPush

Sep 2023 -- Dec 2023

Software Engineer Intern

- Diagnosed DynamoDB hot-partition failure at **9,000+ req/sec**; redesigned partition key strategy - **30% read throughput improvement** under sustained concurrency
- Designed idempotent payment APIs with Redis distributed caching (**85% cache hit rate**) and exponential backoff - enforced exactly-once semantics preventing duplicate transactions under network partitions

PROJECTS

TiMoto AI -- AI Agent Memory Platform

timoto.ai

vLLM, gRPC, Django, PostgreSQL, AWS ECS Fargate, Terraform

- **Sole engineer** - built and operate an AI agent memory platform: session-context persistence layer, vLLM inference engine with agentic workload support, gRPC inter-service APIs, Django REST backend, and AWS cloud infrastructure
- Resolved **production deadlock under concurrent gRPC agent calls** - traced root cause to circular resource acquisition; redesigned call sequencing eliminating circular wait - **100% evaluation success rate at sub-50ms p99**
- Selected **vLLM with PagedAttention** to eliminate KV cache fragmentation under concurrent agent load; deployed with continuous batching - zero OOM failures at production traffic
- Architected multi-AZ ECS Fargate with Terraform IaC and circuit breaker pattern - **99.9% uptime, 44% cost reduction** to \$40-60/month; auto-rollback on health check failure

Open Source Developer Tools -- Pulumi Contributor

[24.4K★github.com/pulumi/pulumi](https://github.com/pulumi/pulumi)

Go, TypeScript, Infrastructure-as-Code (IaC)

- Contributed Go CLI features and bug fixes to Pulumi, a 24.4K-star multi-cloud IaC SDK enabling AWS/Azure/GCP provisioning - under active review by core maintainers
- Analyzed Raft/Paxos consensus in distributed state synchronization layer - studied linearizability guarantees and correctness under concurrent operations and partial failures

EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science · **GPA: 3.75/4.0**

Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures

Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award

SKILLS

Languages: TypeScript, Go, C++, Python, Java, JavaScript, SQL, Bash, Rust, Swift

AI & Agentic: vLLM, PagedAttention, Continuous Batching, Agent memory / session context, LangChain, PyTorch, LLM evaluation, LoRA/QLoRA, Quantization (AWQ/GPTQ)

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Raft/Paxos, Linearizability, Event-driven architecture, Kafka

Cloud & Infrastructure: AWS (ECS Fargate, EC2, S3, RDS), Terraform, Docker, Kubernetes, CI/CD, GitHub Actions, GCP, Azure

Frameworks & Databases: Django, FastAPI, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB